CS 436
Cloud Computing

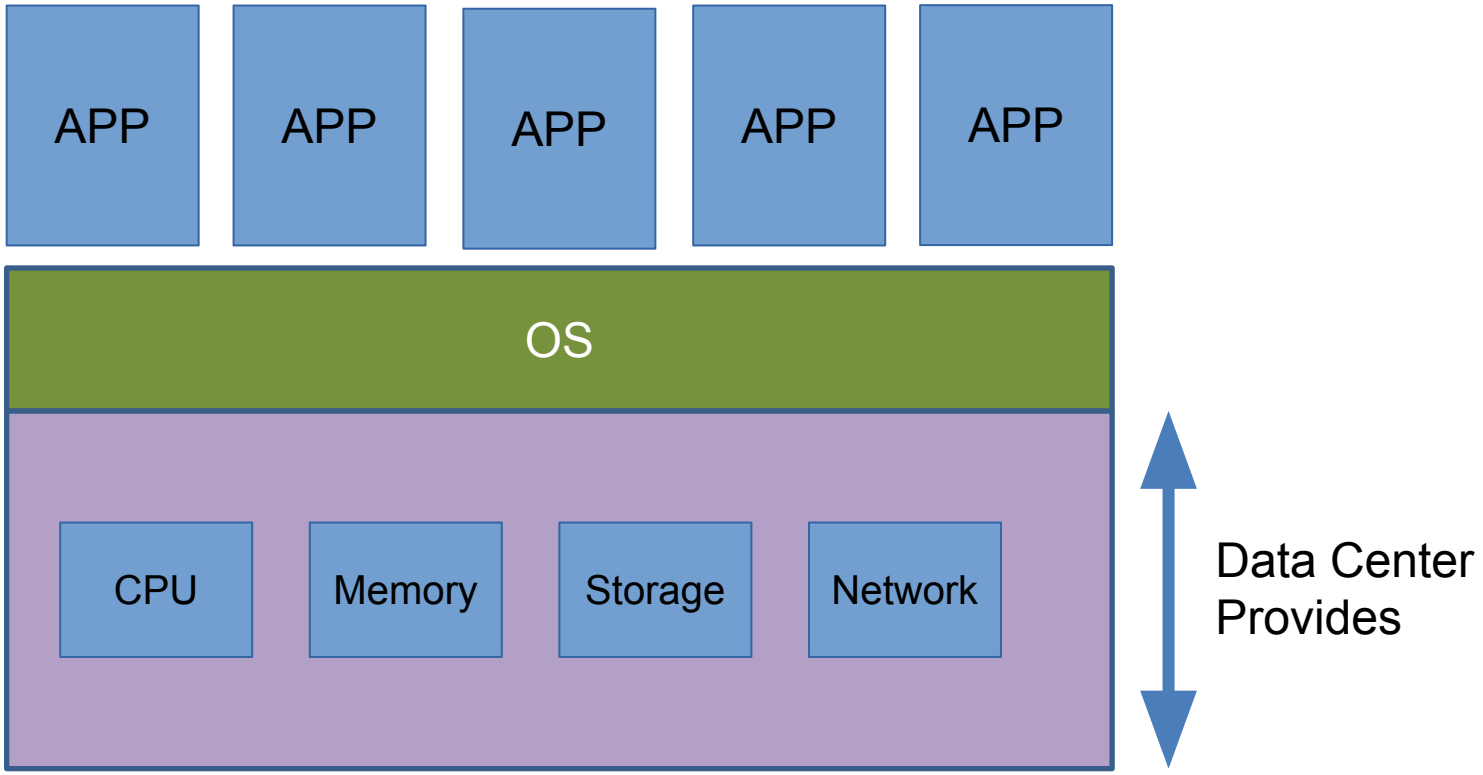
27.02.2025

VERY IMPORTANT

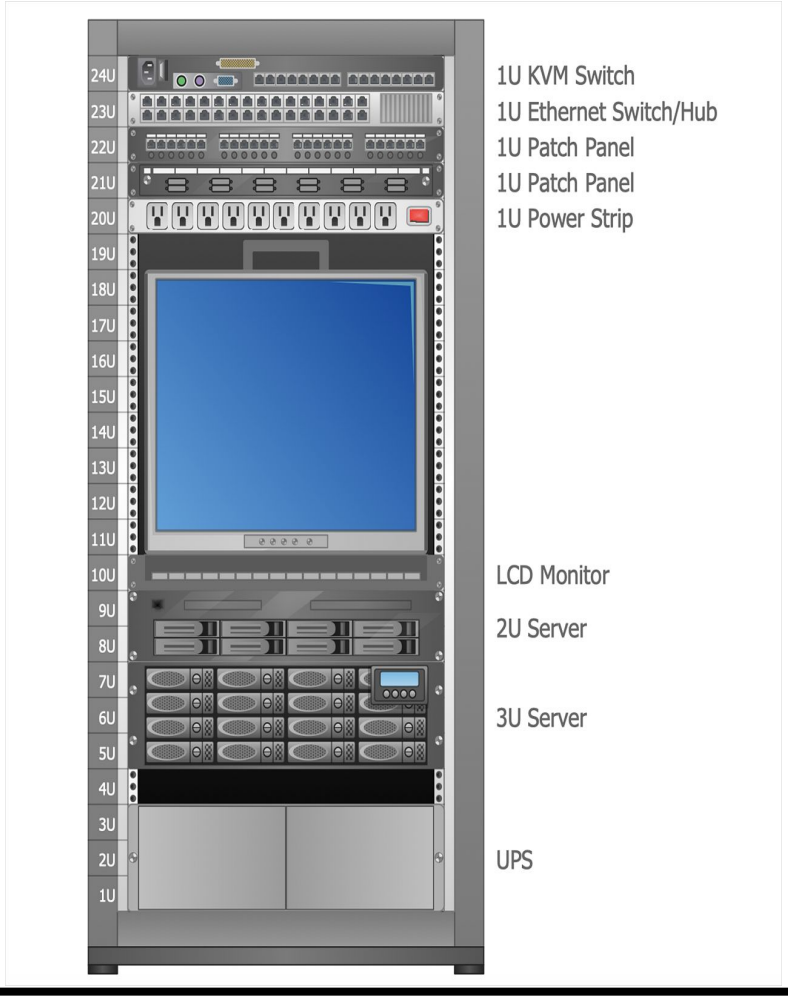


BUDGET ALARM !

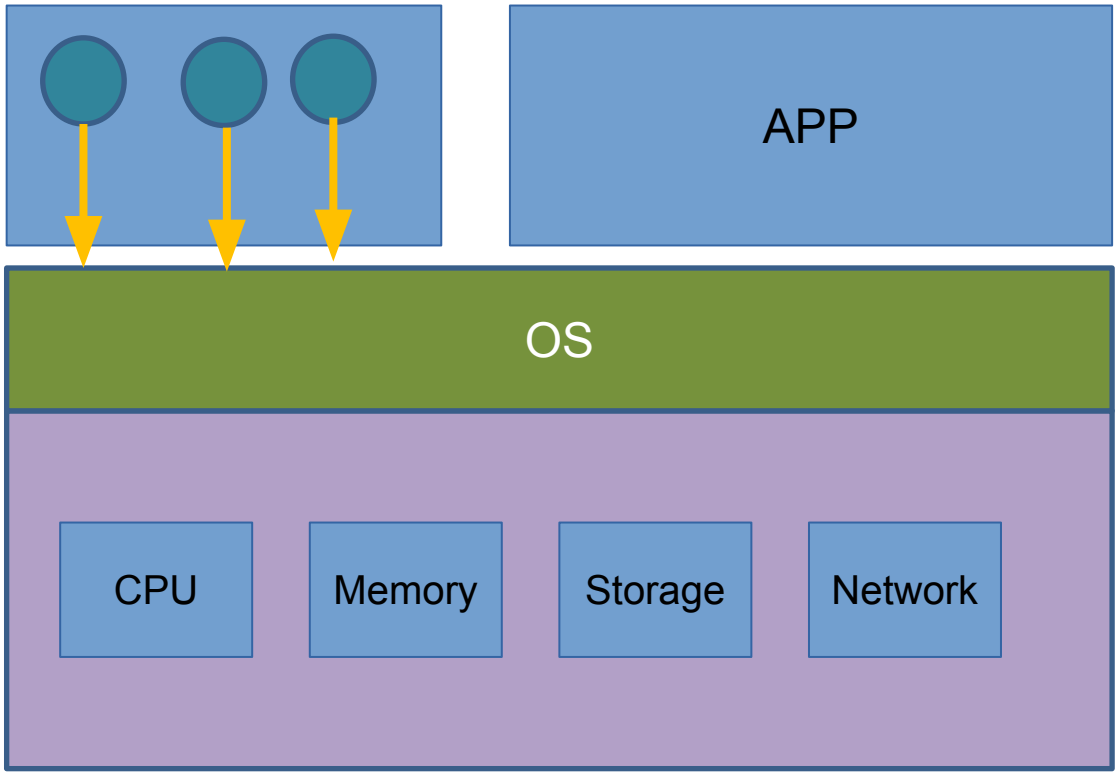
Never operate on AWS, GCP, AZURE without first turning on Budget Alarm.


APP
APP
APP
APP
APP
OS
CPU
Memory
Storage
Network
Data Center
Provides

Rack Server


24U
23U
22U
21U
20U
19U
18U
17U
16U
15U
14U
13U
12U
11U
10U
9U
8U
7U
6U
5U
4U
3U
2U
1U
1U KVM Switch
1U Ethernet Switch/Hub
1U Patch Panel
1U Patch Panel
1U Power Strip
LCD Monitor
2U Server
3U Server
UPS

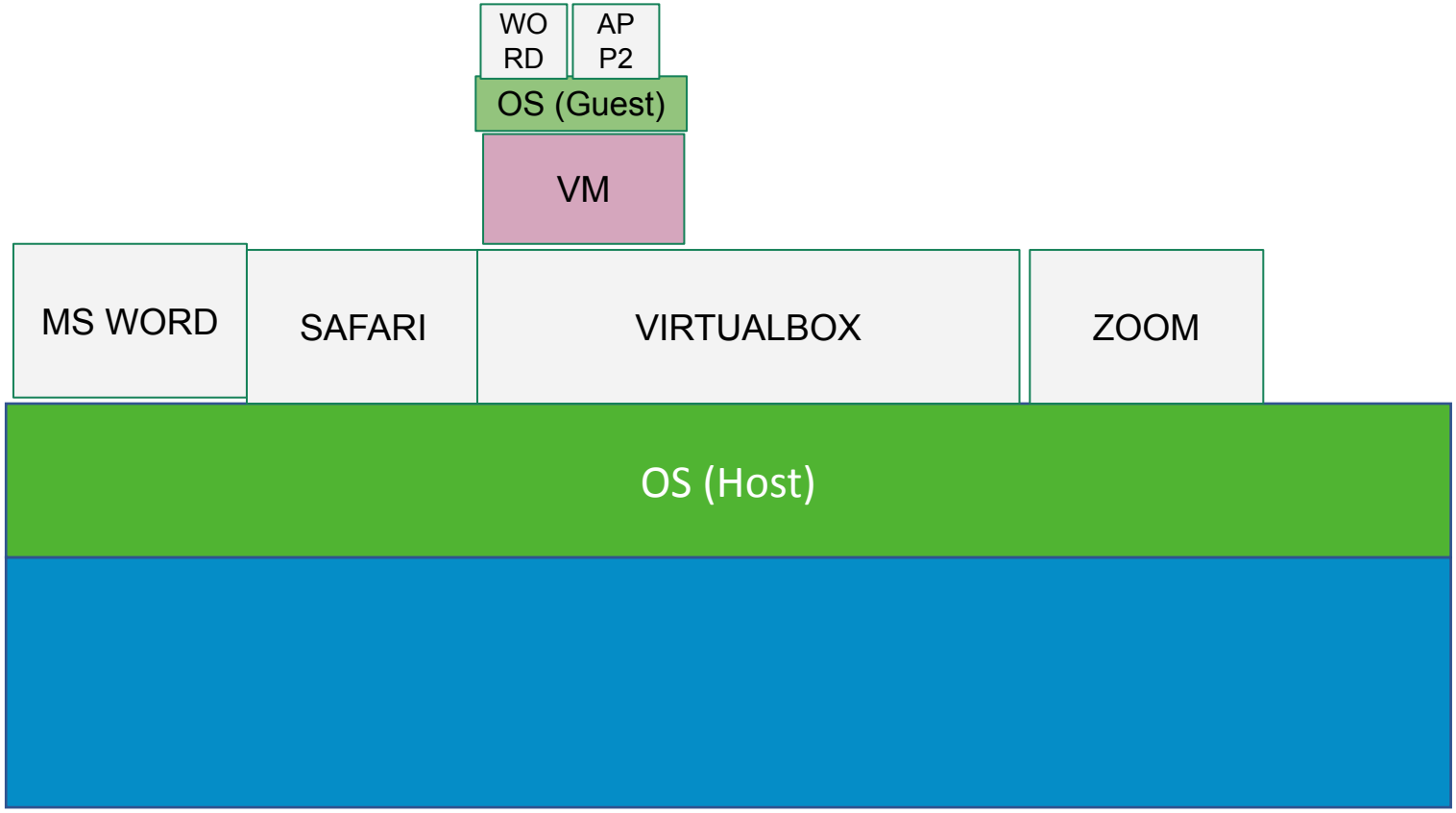
Software runs as a process on OS !



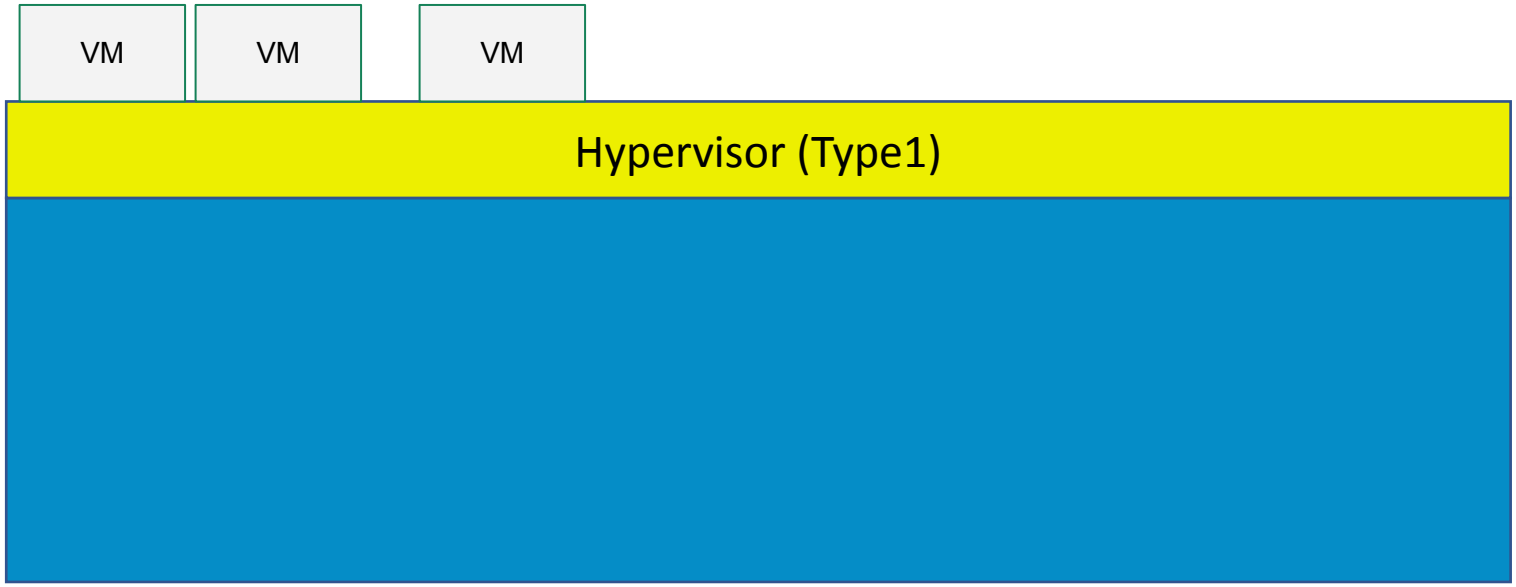
No - Virtualization



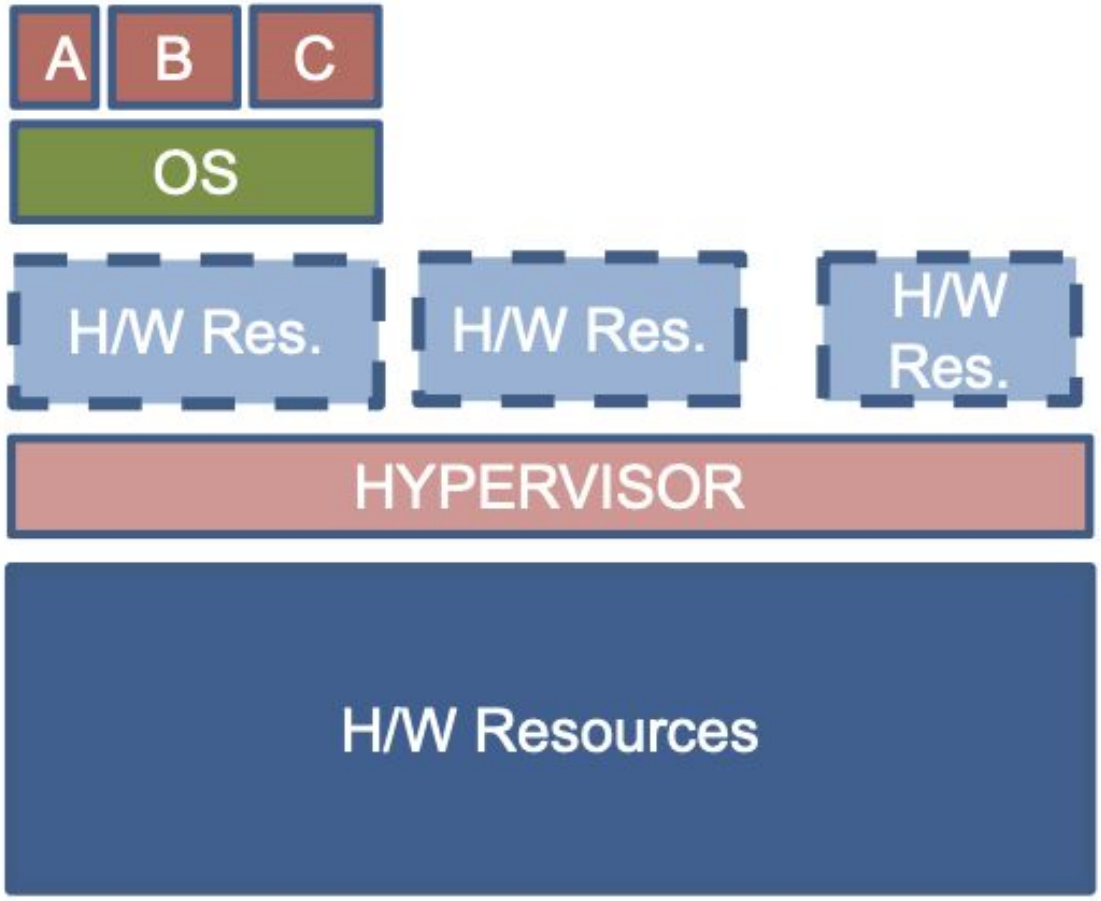
Type 2 Hypervisor Case (Hosted) VirtualBox -VmWare



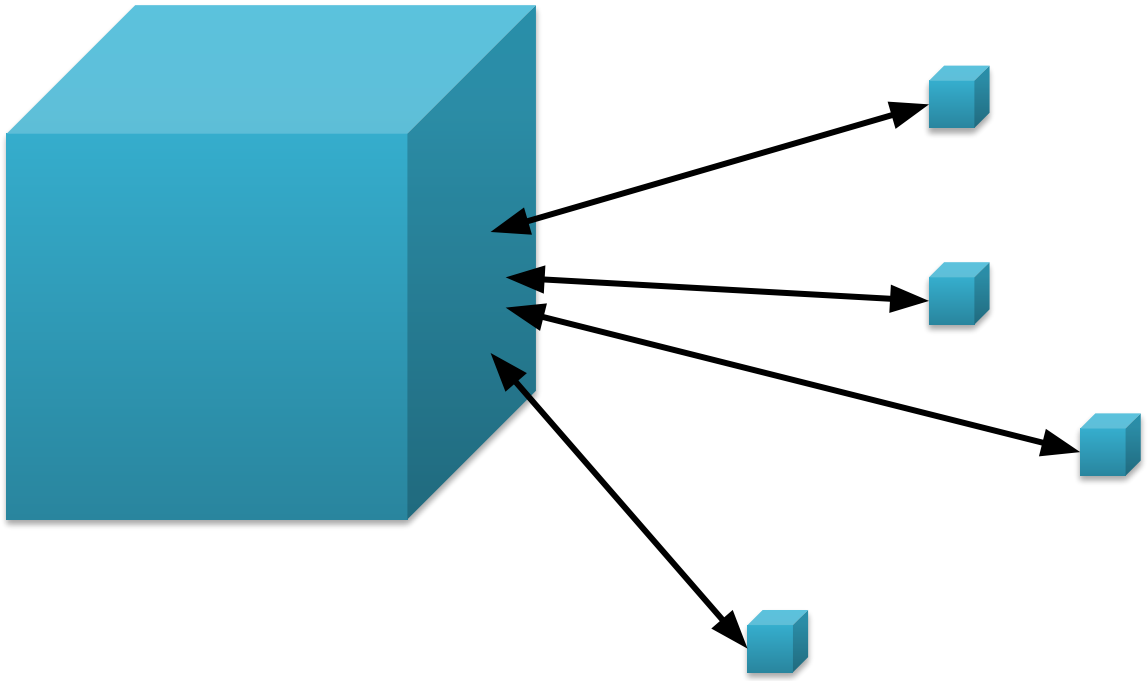
Type 1 Hypervisor Case (bare metal)



In-Class Poll: Is the hypervisor shown in the diagram is Type 1 (Bare Metal) or Type 2(Hosted) ?



Client-Server Architecture



One example: Thin client use case

Diskless Mode



DELL Wyse Z Class Thin Client

Windows Embdedded/Modified Linux , 1.5 GHz single core AMD, 2GB RAM, 4GB Flash

Dimensions: 200 mm x 47 mm x 225 mm
Weight: 1.1. kg

Thin Client

lardware › Computers and Tablets › Thin Clients

Dell Wyse C10LE Thin Client







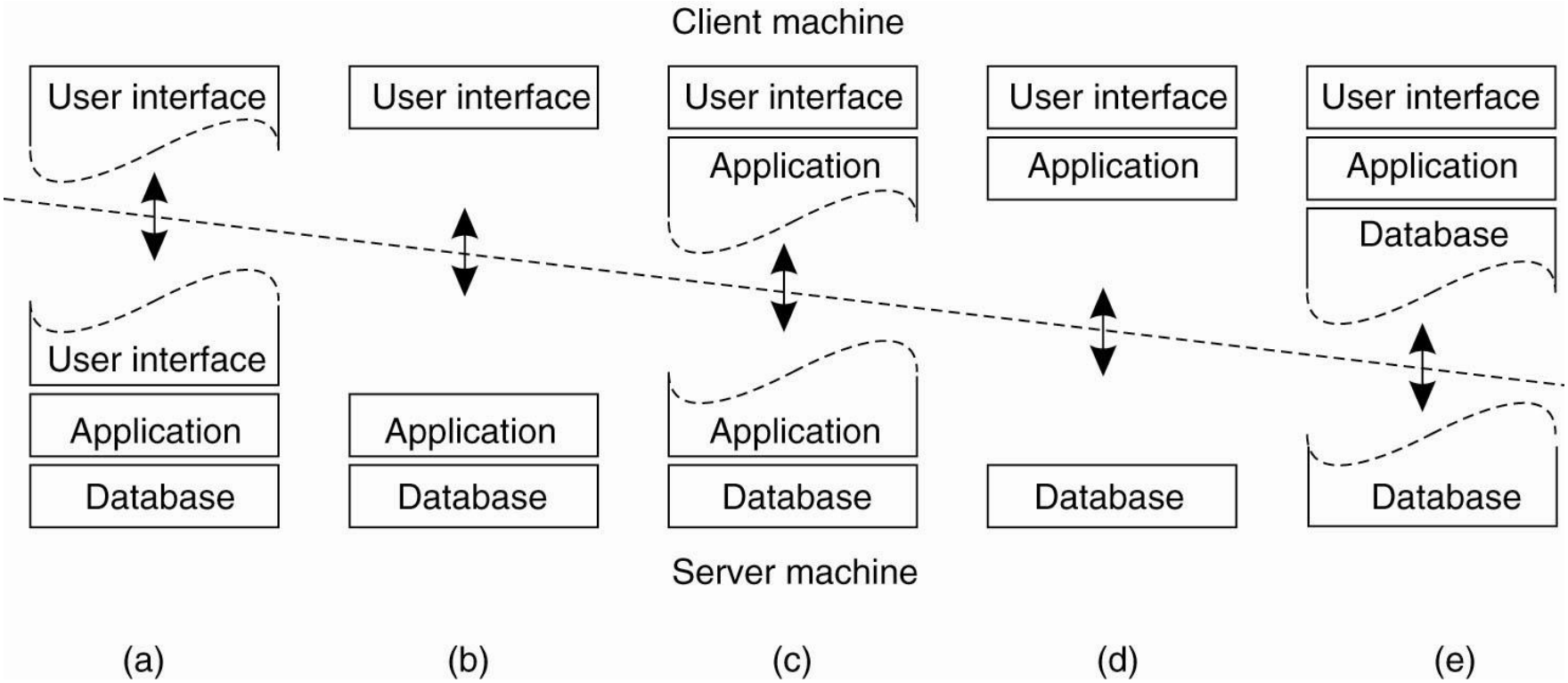

Mfr Part #: 902175-01L
SHI Part #: 21352111

Quick Details Go to Product Specs

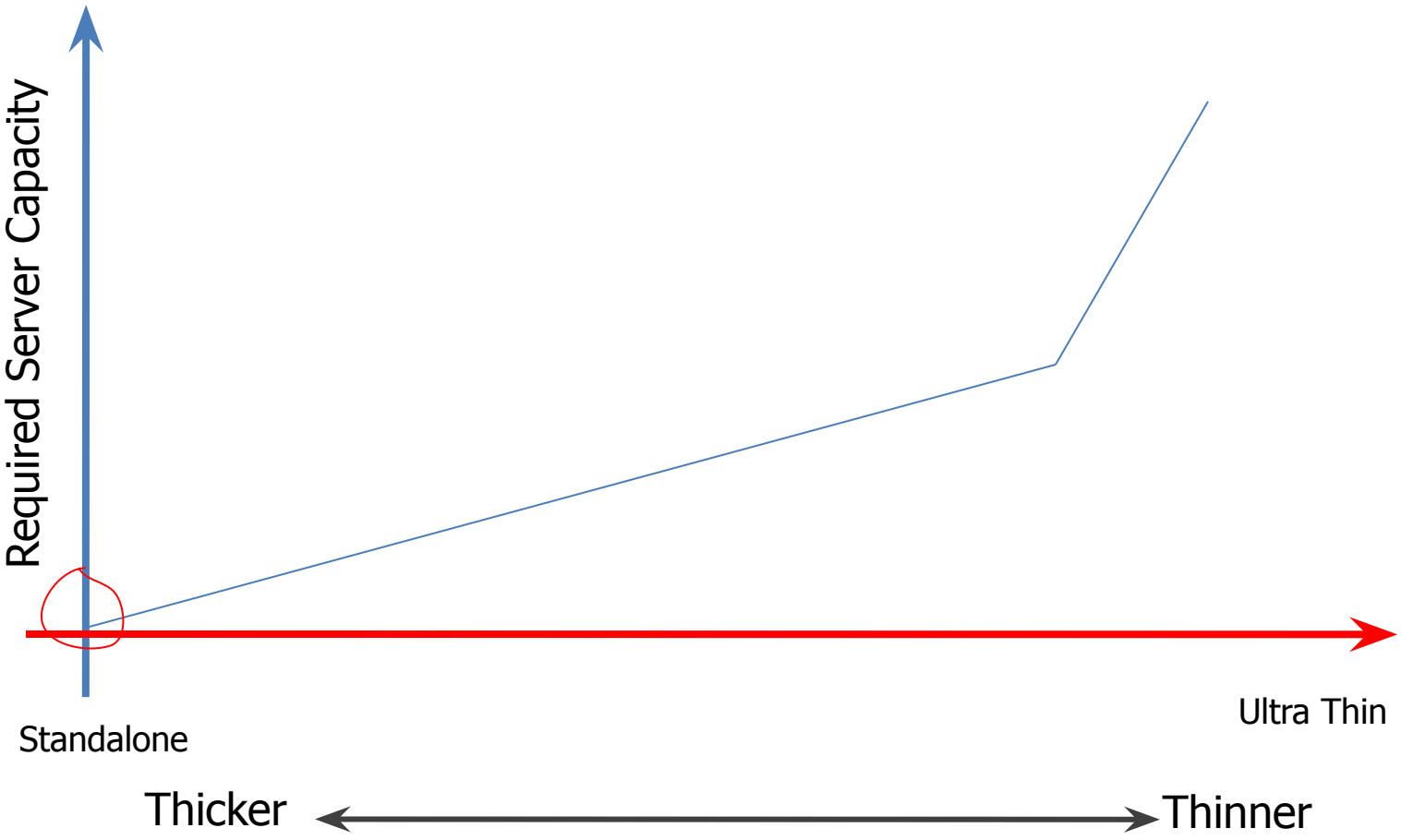
- Thin client
- DTS
- 1 x C7 1 GHz ULV
- RAM 512 MB
- flash 128 MB

1 Add to Cart

Functionality Assignment in C/S Architecture

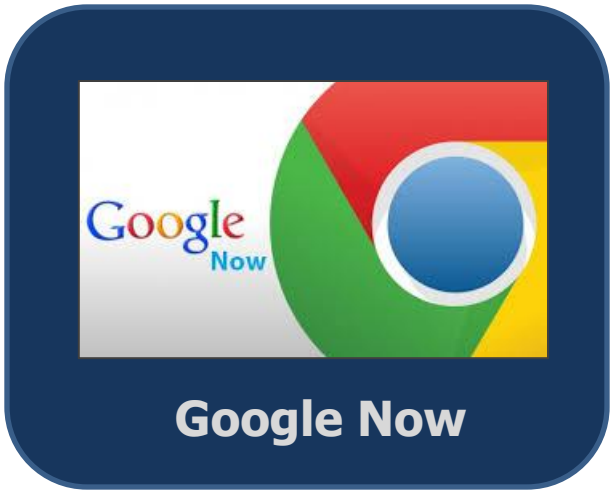


Thin vs. Thick Clients



Intelligent Personal Assistant Software

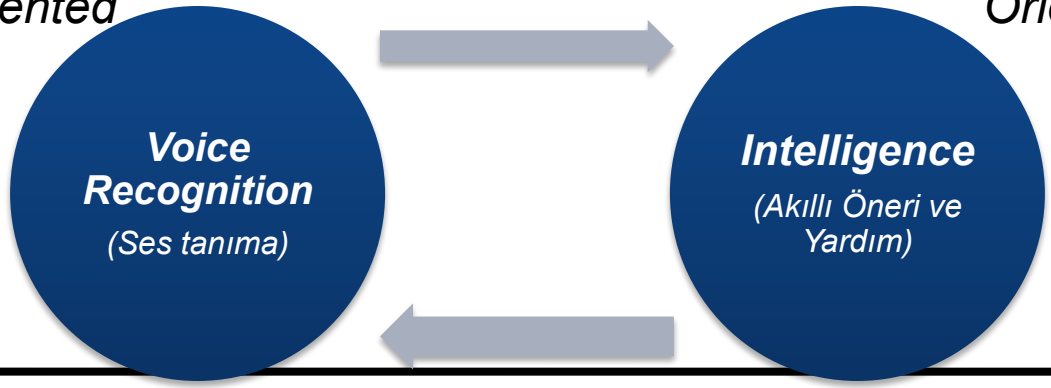
(Akıllı Kişisel Yardımcı Uygulamaları)



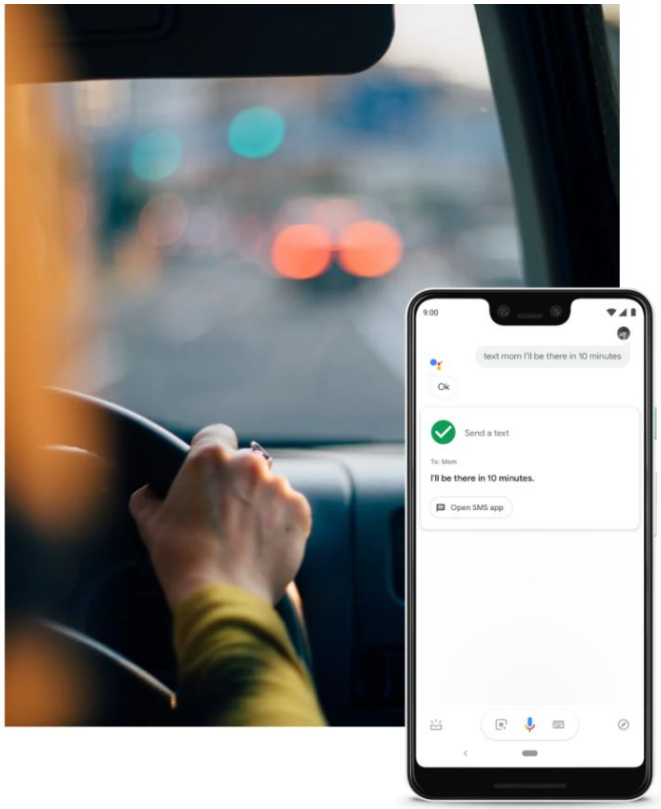
"Voice Search" Oriented



Personal Assistance Oriented



Make Google text it.



Hey Google, text Mom I'll be there in 10 minutes

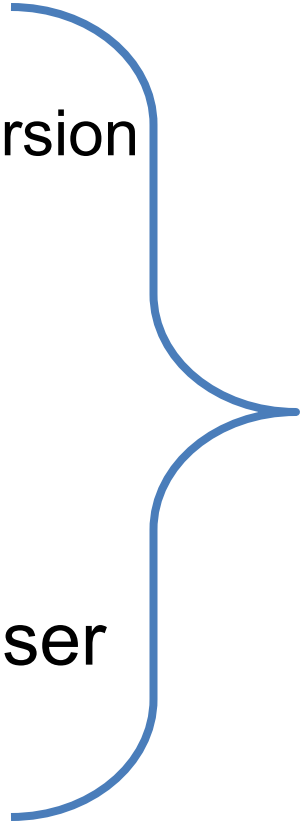
Learn more

Google Now is retired !

Discover what Google Assistant is

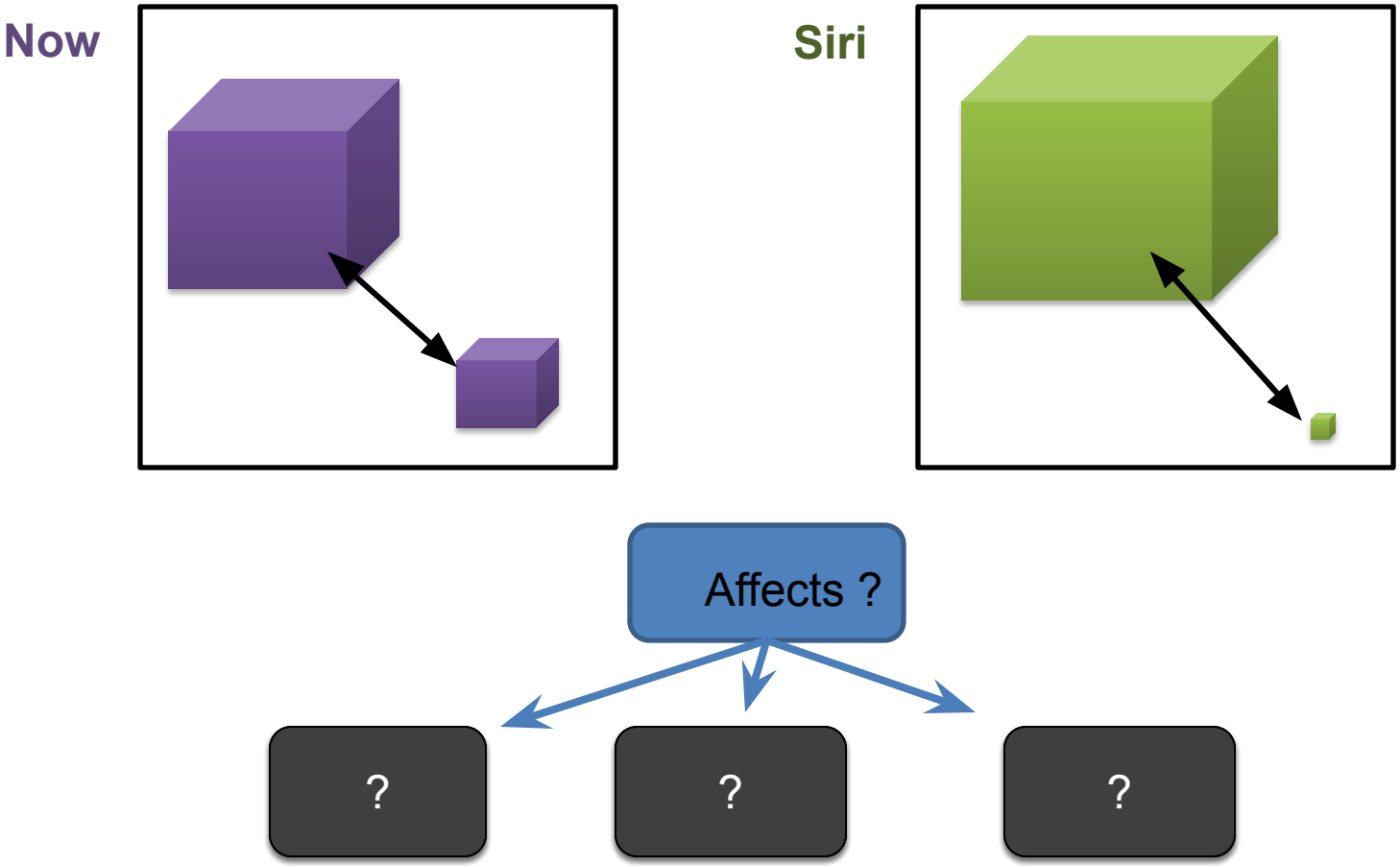
Functinality & Network Communication

- Speech processing
  - Analog-Digital conversion
  - Noise removal
- Speech to text
- Text interpretation
- Answer / reply
- Presenting it to the user

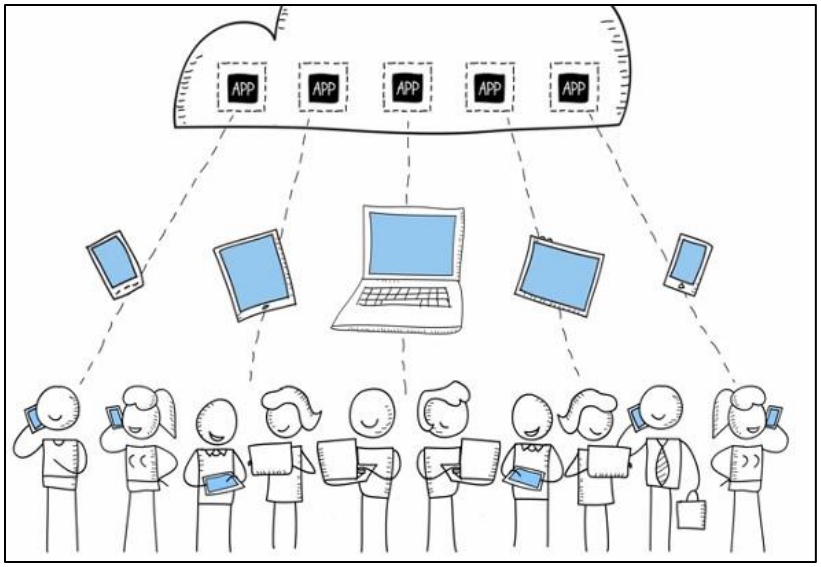


How much of it runs on handset ? How much of it runs on cloud?

Guessing the Server Load



Case Study: Amazon Appstream

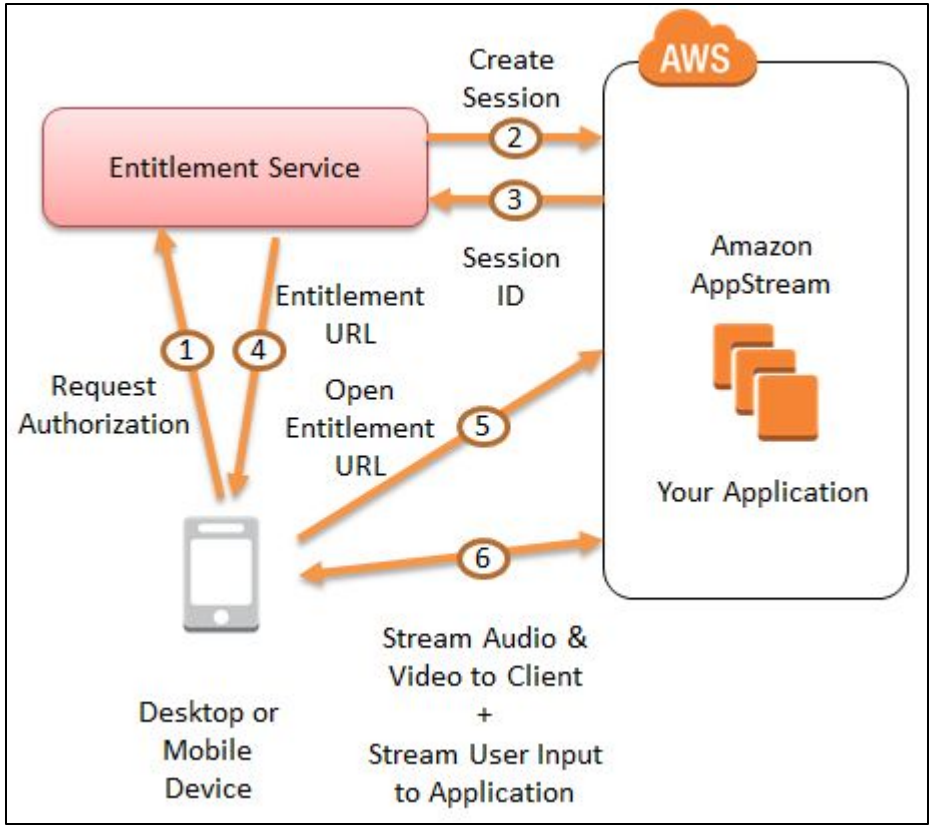


AIM: Stream resource intensive applications (games) from the cloud

Run the application on cloud infrastructure

LATENCY !

Case Study: Amazon Appstream



Use the Amazon AWS infrastructure

Propritery STX protocol allows dynamic adjustment of encoding.

Load Balancing

- DNS based sol’ns
- HTTP redirection
- Load Balancer
  - Layer 4 Load Balancer (Legacy Load balancing)
  - Layer 7 Load Balancer (Layer 7 Switching)

DNS Based Load Balancing

Name	TTL	Type	IP
www.example.com.	10800	A	192.168.1.2
www.example.com.	10800	A	10.2.54.4

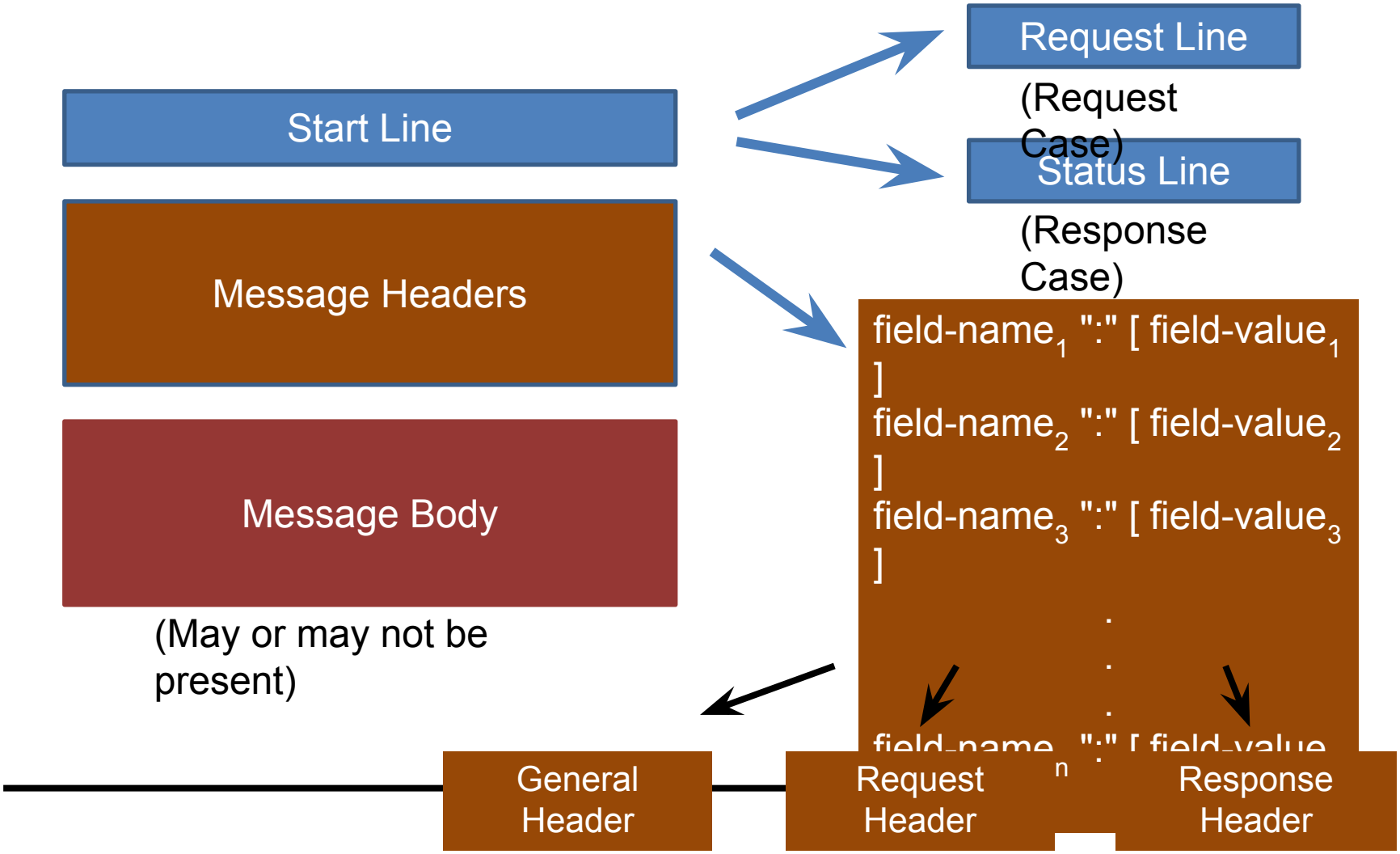
Round robin policy supported by DNS servers

Ref:
http://oldcp.dnsmadeeasy.com/enterprisedns/records.html

HTTP Redirection

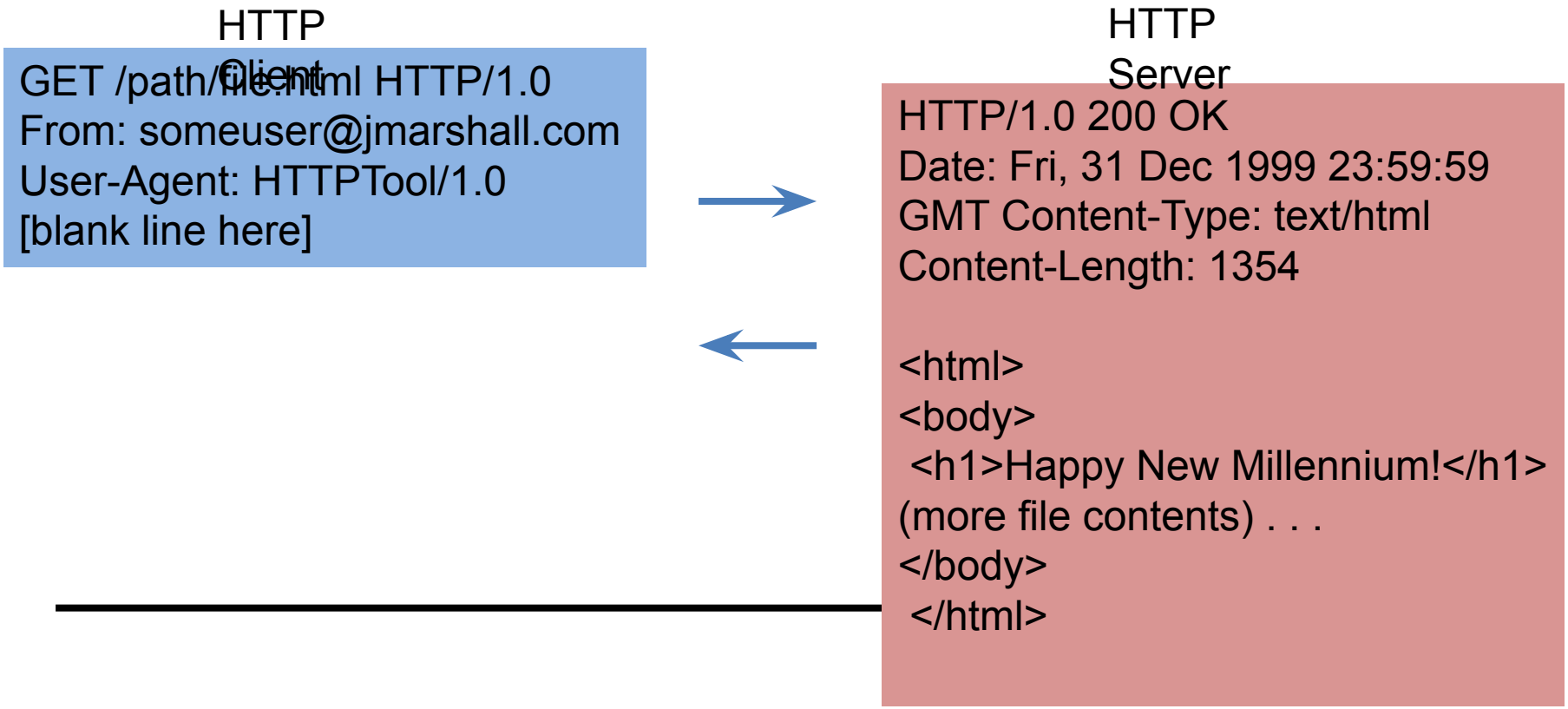
- Advantages: Very simple to implement
- Weaknesses
  - Internals of the server organization becomes visible to the client. Access Transparency violated -> Possible security problems
  - Does not enhance availability: Clients do not automatically switch to another replica in case of failure
  - Scalability: Depends on the delivery speed of the initial document.

Generic HTTP Message



Typical Messaging Sequence

- We want to access:
  http://www.somehost.com/path/file.html



Response Classes (RFC2616)

- 1xx: Informational - Request received, continuing process
- - 2xx: Success - The action was successfully received, understood, and accepted
- - 3xx: Redirection - Further action must be taken in order to complete the request
- - 4xx: Client Error - The request contains bad syntax or cannot be fulfilled
- - 5xx: Server Error - The server failed to fulfill an apparently valid request

Yahoo Redirection

For more information on HTTP see **RFC 2616**

HTTP(S)-URL: http://www.yahoo.com

HTTP version: HTTP/1.1 HTTP/1.0 (with Host header) HTTP/1.0 (without Host header)

Raw HTML view Accept-Encoding: gzip • Request type: GET POST HEAD TRACE

User agent: Web-Sniffer

Flattr 155

HTTP Request Header

```
Connect to 87.248.112.181 on port 80 ... ok

GET / HTTP/1.1[CRLF]
Host: www.yahoo.com[CRLF]
Connection: close[CRLF]
User-Agent: Web-sniffer/1.0.44 (+http://web-sniffer.net/)[CRLF]
Accept-Encoding: gzip[CRLF]
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8[CRLF]
Accept-Language: en-GB,en-US;q=0.8,en;q=0.6,tr;q=0.4[CRLF]
Accept-Charset: ISO-8859-1,UTF-8;q=0.7,*;q=0.7[CRLF]
Cache-Control: no-cache[CRLF]
Referer: http://web-sniffer.net/[CRLF]
[CRLF]
```

Yahoo Redirection - II

HTTP Response Header

Name	Value
Status: HTTP/1.1 302 Found	
Date:	Tue, 21 May 2013 12:00:46 GMT
P3P:	policyref="http://info.yahoo.com/w3c/p3p.xml", CP="CAO DSP COR CUR ADM DEV TAI PSA PSD IVAi IVDi CONi T
Cache-Control:	private
X-Frame-Options:	SAMEORIGIN
Set-Cookie:	fpc=d=GM9G2jOUTowmMDxTEZqMwNkz7vR8j_iZCWLz93qFqrHHpl0n.mVPm1UslQRCE2gXuBL81_4biWISePOkM0QAr &v=2; expires=Wed, 21-May-2014 12:00:46 GMT; path=/; domain=www.yahoo.com
Location:	http://de.yahoo.com/?p=us
Vary:	Accept-Encoding
Content-Type:	text/html; charset=utf-8
Age:	0
Connection:	close
Server:	YTS/1.20.13

Commercial Load Balancer Spec





	ALB-X E4	ALB-X E5
Chassis	1U Half Depth	1U Full Depth
CPU	Quad Core	2 X Hex Core
Memory	8GB	12GB - 64GB
Network Interface	4x 10/100/1000	8x 10/100/1000
Max Interface	8x 10/100/1000	16x 10/100/1000 or 2 x 10GbE
Power Consumption	71W to 100W maximum	150W to 200W maximum
HTTP Max Throughput	8 Gbps	25 Gbps
Layer 7 – HTTP Requests Per Second	70,000 rps	400,000 rps
SSL Transaction Per Second	6000 tps	20,000 tps

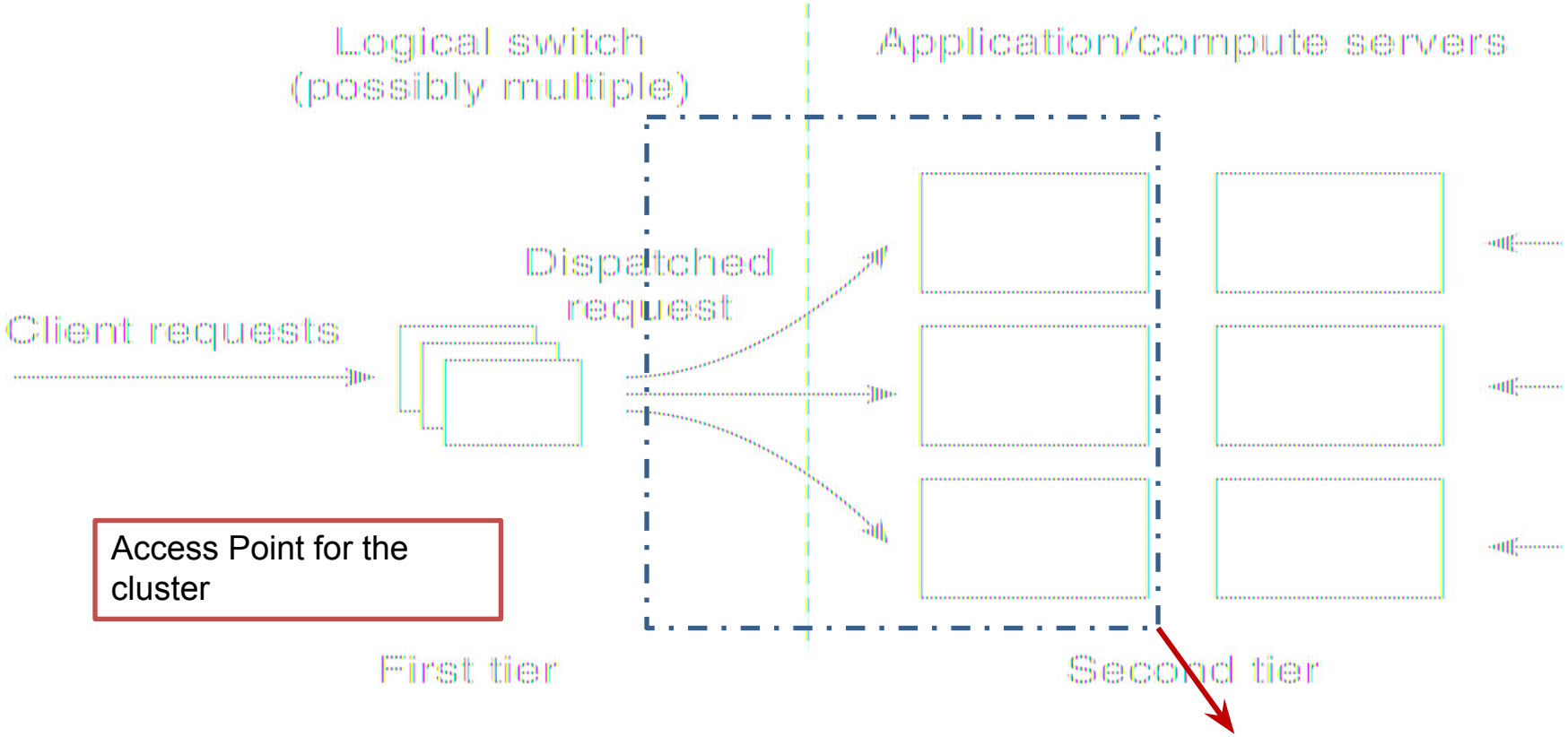
Scheduling Methods (Layer 4 Balancer)

- Round Robin
- Weighted Round Robin
- Least Connection
- Weighted Least Connection

Layer 4 vs. Layer 7 Load Balancing

- İdentical functionality should be served in L4 Load Balancing (No dedicated PHP, image etc. server can exist)
- L7 load balancing allows partial replication and dedicated servers.

Load Balancing



Homogeneous vs. Heterogeneous Application Servers

HAProxy: Open Source Load Balancer

HAProxy

The Reliable, High Performance TCP/HTTP Load Balancer

HAPROXY COMMUNITY EDITION

Mirror Sites: Master
Language: English

Quick links

- **Quick News**
- Recent News
- Description
- Main features
- Supported Platforms
- Performance
- Reliability
- Security
- **Download**
- **Documentation**
- **Live demo**
- **They use it!**
- Enterprise Features
- Third party extensions
- **Commercial Support**
- Add-on features
- Other Solutions
- Contacts
- External links
- Discussions
- Slack channel
- **Mailing list**
- **10GbE load-balancing**
- **Contributions**
- **Coding style**
- **Open Issues**
- **Known bugs**

- **HATop: Ncurses Interface**
- **Herald: load feedback agent**
- **haproxystats: stats collection**
- **Alpine-based Docker images**
- **Debian-based Docker images**
- **RHEL-based Docker images**
- **Debian/Ubuntu packages**

Latest versions

Branch	Release date	End of life	Latest version	Changelog	Links
2.5-dev	~2021-Q4	2023-Q1 (dev » stable)	2.5-dev13	2021/11/06	git / web / dir / announce
2.4	2021-05-14	**2026-Q2 (LTS)**	2.4.8	2021/11/03	git / web / dir / announce / bugs
2.3	2020-11-05	2022-Q1 (stable)	2.3.15	2021/11/04	git / web / dir / announce / bugs
2.2	2020-07-07	**2025-Q2 (LTS)**	2.2.18	2021/11/05	git / web / dir / announce / bugs
2.0	2019-06-16	**2024-Q2 (LTS)**	2.0.25	2021/09/07	git / web / dir / announce / bugs
1.8	2017-11-26	**2022-Q4 (critical fixes only)**	1.8.30	2021/04/12	git / web / dir / announce / bugs
1.7	2016-11-25	2021-Q4 (critical fixes only)	1.7.14	2021/03/31	git / web / dir / announce / bugs

Hide/Show unmaintained

Quick News

November 5th, 2021 : Cool hardware donation from Zerodha

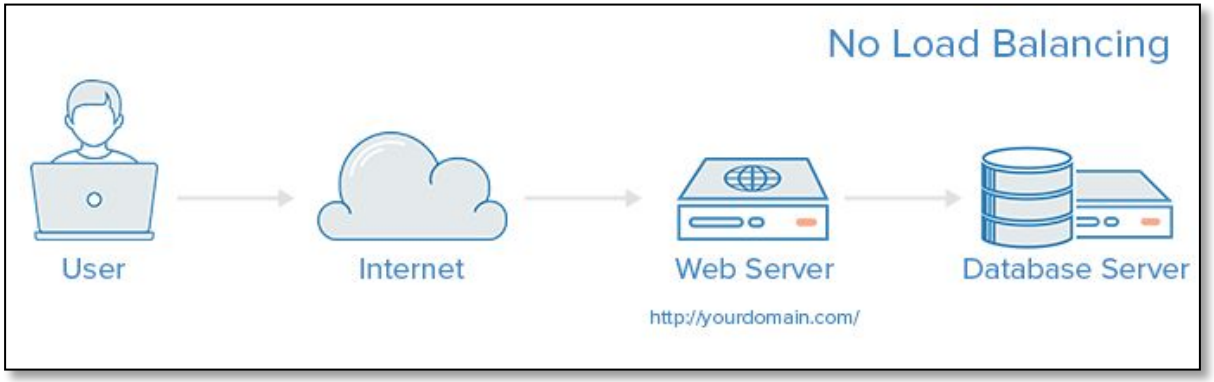


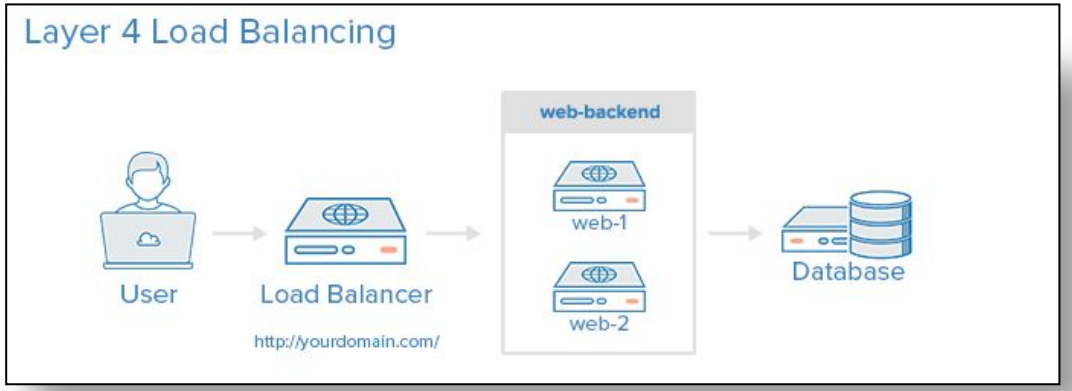
Just after the HAProxy 2.4.0 release in May this year, Kailash Nadh, CTO of Zerodha contacted me and offered to donate a pair of SolidRun HoneyComb LX2 boards to help us continue to improve out threading scalability. With 16 ARM cores, 32GB RAM, 64GB eMMC, 4x10GbE, and PCI-e in a tiny and quiet form factor, there definitely is plenty of potential to emphasize the cost of contention and to further improve our data model and our algorithms! As usual with SolidRun, the hardware design is really awesome (it feels like you're installing a PC, and they're about the only ones featuring the console port). However I must say I was really disappointed by their complete absence of communication during the 5 months it took to ship the board. Vishnu of Zerodha and me probably asked them for status updates 8 times in total and each time we were told "we'll contact you when it ships, no need to ask all the time". Last time we were told the parcel had been sent two weeks ago to DHL who had no way to contact me... no comment! Emailing a tracking number doesn't cost much, especially when their own tracking site still shows the order as "processing" after I received it! Hopefully SolidRun manages to address this deficiency in the future because right now they're among the very few who make affordable server-grade development boards and I hope to continue to be able to order their great products without trouble... Anyway, more news later when the tests really begin! For now the boards boot and are fast (hint: haproxy-2.5 builds in 6 sec). Many big thanks to Kailash and Vishnu of Zerodha for this very kind contribution!

November 3rd, 2021 : HAProxyConf 2021 is in less than two weeks!

The conference is still scheduled for November 16-17 and will be free to watch on https://www.haproxyconf.com/, with live Q&A sessions after each talk. The list of speakers and their presentations is already published. The conference schedule is being finalized and is to be published in the forthcoming days, please check the site to stay up to date, or register to be notified about last-minute updates. See you soon!

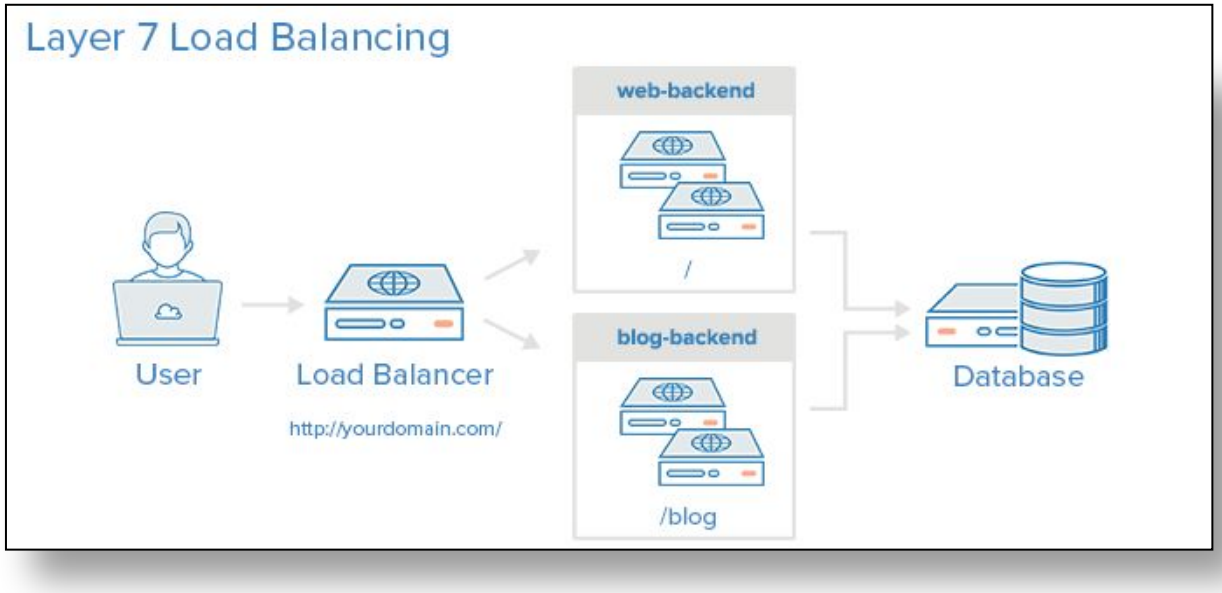
Web Load Balancing





https://www.digitalocean.com/community/tutorials/an-introduction-to-haproxy-and-load-balancing-concepts

Layer 7 Load Balancing



Load Balancing Features: GCP Case

- **Single anycast IP address.** With Cloud Load Balancing, a single anycast IP address is the frontend for all of your backend instances in regions around the world. It provides cross-region load balancing, including automatic multi-region failover, which moves traffic to failover backends if your primary backends become unhealthy. Cloud Load Balancing reacts instantaneously to changes in users, traffic, network, backend health, and other related conditions.
- **Software-defined load balancing.** Cloud Load Balancing is a fully distributed, software-defined, managed service for all your traffic. It is not an instance-based or device-based solution, so you won't be locked into physical load balancing infrastructure or face the HA, scale, and management challenges inherent in instance-based load balancers.
- **Seamless autoscaling.** Cloud Load Balancing can scale as your users and traffic grow, including easily handling huge, unexpected, and instantaneous spikes by diverting traffic to other regions in the world that can take traffic. Autoscaling does not require pre-warming: you can scale from zero to full traffic in a matter of seconds.
- **Layer 4 and Layer 7 load balancing.** Use Layer 4-based load balancing to direct traffic based on data from network and transport layer protocols such as TCP, UDP, ESP, or ICMP Use Layer 7-based load balancing to add request routing decisions based on attributes, such as the HTTP header and the uniform resource identifier.
- **External and internal load balancing.** You can use external load balancing when your users reach your applications from the internet and internal load balancing when your clients are inside of Google Cloud.
- **Global and regional load balancing.** Distribute your load-balanced resources in single or multiple regions, to terminate connections close to your users, and to meet your high availability requirements.

REF: https://cloud.google.com/load-balancing/docs/load-balancing-overview